

Synopsis

This portfolio analysis evaluates the weekly returns and risk characteristics of Shell plc and Exxon Mobil and compares them with the market benchmark represented by the S&P 500. Both companies operate in the global energy sector and are frequently compared by investors due to their similar industry exposure. The goal of this analysis is to examine their returns, volatility, and risk-adjusted performance in order to evaluate their suitability for investment portfolios.

The analysis is based on weekly returns from March 2025 to March 2026. The average weekly return for Shell was 0.52%, while Exxon Mobil had a slightly higher mean return of 0.70%, indicating better average performance. In terms of volatility, Shell had a sample variance of 0.09% and a standard deviation of 3.11%, whereas Exxon Mobil had a variance of 0.11% and a standard deviation of 3.34%, suggesting that Exxon Mobil's returns were slightly more volatile.

The spread of returns was evaluated using maximum, minimum, and range values. Shell's highest weekly return was 6.64% and its lowest return was -11.83%, giving a range of 18.47%. Exxon Mobil had a maximum return of 7.53% and a minimum of -11.38%, resulting in a range of 18.91%. To detect unusual observations, the interquartile range (IQR) method was used. Shell had Q1 of -0.74% and Q3 of 2.42%, producing an IQR of 3.16%. Exxon Mobil had Q1 of -0.91% and Q3 of 2.73%, giving an IQR of 3.65%. The calculated fences for Shell were 7.17% and -5.49%, while for Exxon Mobil they were 8.21% and -6.39%, identifying extreme negative returns as outliers.

Risk-adjusted performance was evaluated using the Sharpe ratio with a weekly risk-free rate of 0.07%. Shell had a Sharpe ratio of -2.23, while Exxon Mobil had a ratio of 0.21, indicating that Exxon Mobil provided better compensation for risk during the sample period. The covariance between the two stocks was 0.0007067, and the correlation coefficient was 0.697, indicating a strong positive relationship between their returns.

A regression analysis was also conducted to evaluate the relationship between Shell's returns and the overall market. The regression coefficient was 0.289, suggesting that a 1% increase in the S&P 500 is associated with an average 0.29% increase in Shell's returns. However, the R^2 value of 0.046 shows that only 4.6% of Shell's return variation is explained by market movements, and the p-value of 0.126 indicates that the relationship is not statistically significant at the 5% level.

Overall, Exxon Mobil demonstrated slightly higher average returns and slightly greater volatility than Shell, while both companies showed similar return patterns and a strong positive correlation due to their exposure to the same industry conditions.